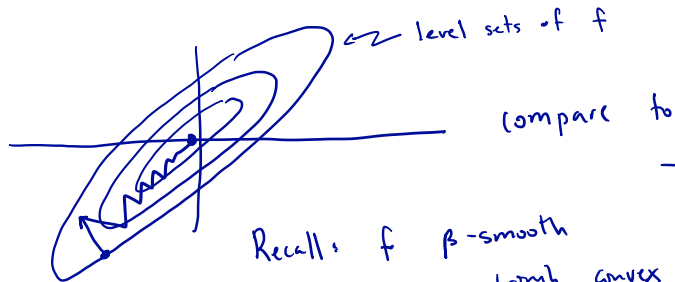


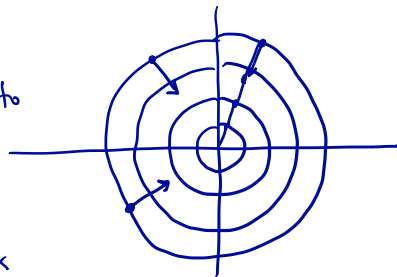
Newton's Method - a second view

Gradient descent for ill conditioned quadratic

$$\text{GD: } x_{t+1} = x_t - \eta \nabla f(x_t)$$



compare to



Recall: f β -smooth
 α -strongly convex

$$\text{GD: } \underline{O\left(\frac{\beta}{\alpha} \log(1/\epsilon)\right)}$$

Gradient descent after affine transformation

min: $f(x)$ β -smooth. α -strongly convex

Rescale space: $x = Ay$, A psd & invertible
 $y = A^{-1}x$

Define: $g(y) = f(Ay)$

min: $g(y)$
 y
 y^*

min: $f(x)$
 x
 x^*

$g(y^*) = f(x^*)$
 $x^* = Ay^*$
 $A^T x^* = y^*$

Q: If using GD as minimization alg., is it equally hard/easy to minimize $g(y)$, $f(x)$? Or could one possibly be easier?

Gradient descent after affine transformation: Example

$$f(x) = \frac{1}{2} x^T Q x, \quad Q = \begin{bmatrix} 100 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\beta = 100, \quad \alpha = 1 \rightarrow \kappa_f = \beta/\alpha = 100$$

$$\text{Let } A = \begin{bmatrix} 1/10 & 0 \\ 0 & 1 \end{bmatrix} \quad x = Ay$$

$$\begin{aligned} g(y) &= f(Ay) = \frac{1}{2} y^T A^T Q A y \\ &= \frac{1}{2} y^T \begin{bmatrix} 1/10 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 100 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1/10 & 0 \\ 0 & 1 \end{bmatrix} y \\ &= \frac{1}{2} y^T \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} y = \frac{1}{2} y^T y. \end{aligned}$$

g is 1-smooth and 1-strongly convex.

$$\alpha = \beta = 1, \quad \kappa_g = 1.$$

$$\text{From any } y_0 : \quad y_1 = y_0 - 1 \cdot \nabla \left(\frac{1}{2} y_0^T y_0 \right) = y_0 - y_0 = 0 = y^*$$

$\Rightarrow$ GD needs 1 step to converge.

GD for f :

Say $x_0 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

$$x_1 = x_0 - \frac{1}{100} \nabla \left(\frac{1}{2} x_0^T Q x_0 \right)$$
$$= x_0 - \frac{1}{100} (Q x_0)$$

$$\begin{pmatrix} 100 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$x_1 = x_0 - \frac{1}{100} \cdot \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ .99 \end{pmatrix} \rightarrow \text{not optimal.}$$

in step...

GD: not invariant to affine transformation

$$f(x), \quad g(y) = f(Ay) \quad \text{using} \quad x = Ay.$$

$$\begin{array}{ccc} x_0 & \longleftrightarrow & y_0 = Ax_0 \\ x_1 & & y_1 \\ \vdots & & \vdots \\ x_t & & y_t \\ \vdots & & \vdots \\ x^* & \longleftrightarrow & y^* = Ax^* \end{array}$$

What we have learned: affine transformations help

(A) GD not inv. to affine tr.

(B) Aff. transformations have the potential

to greatly speed up GD by decreasing
 $k = \beta/\alpha$.

(A) + (B) : Idea: Let's combine GD with the
right/best affine transformation at every step.

→ i.e., find the transformation that locally gives us
condition number ≈ 1 .

What is a good transformation?

$y = Ax$ so that locally $g(y)$ has condition # = 1.

$$\text{Local condition \# : } \lambda_{\max}(\nabla^2 f(x)) / \lambda_{\min}(\nabla^2 f(x))$$

$\Rightarrow$ need A so that: for $g(y) = f(Ay)$ ($x = Ay$)

Work backward
to compute A

$$\nabla^2 g(y) = I$$

$$g(y) = f(Ay)$$

$$\nabla g(y) = A^T \nabla f(Ay)$$

$$\nabla^2 g(y) = A^T (\nabla^2 f(Ay)) A$$

$$\begin{aligned} \text{Take } A &= (\nabla^2 f(Ay))^{-1/2} \\ \Rightarrow \nabla^2 g(y) &= (\nabla^2)^{-1/2} \nabla^2 (\nabla^2)^{-1/2} \\ &= I \end{aligned}$$

Fact: If a matrix M is sym. with +ve e-values
then $M^{-1/2}$ is a sym. " " "

Summary: Newton = GD + transformation at each step

GD using this transformation?

$$y_{t+1} = y_t - \eta \nabla g(y_t), \quad y_t = Ax_t, \quad A = \left(\nabla^2 f(x_t) \right)^{-1/2}$$

$$x_{t+1} = Ay_{t+1}$$

$$y_{t+1} = A^{-1}x_{t+1} = A^{-1}x_t - \eta A^T \nabla f(x_t)$$

multiply by A:

$$\Rightarrow \boxed{x_{t+1} = x_t - \eta (AA^T) \nabla f(x_t)}$$
$$\Rightarrow \boxed{x_{t+1} = x_t - \eta \left(\nabla^2 f(x_t) \right)^{-1} \nabla f(x_t)}$$

Newton's Method.

$\Rightarrow$ Newton = GD + locally optimal/best transformation at each step.